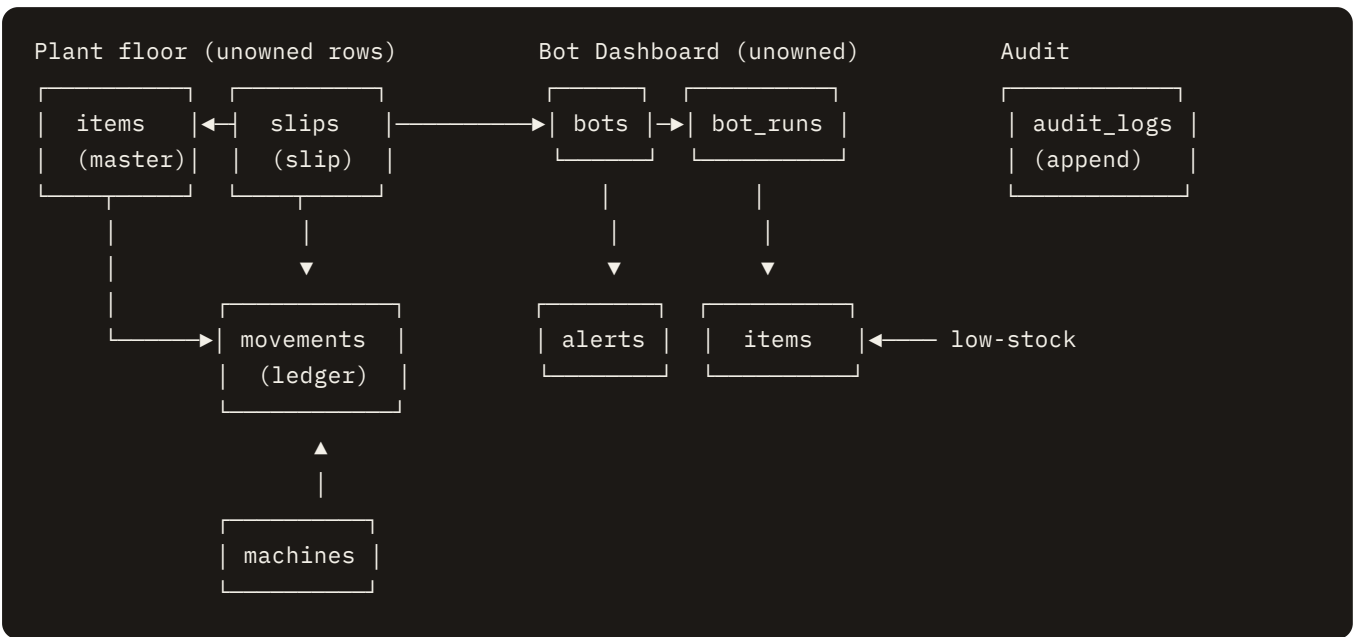




Database Design — Procurement Hub & Bot Dashboard

Single source of truth: migrations/*.sql (applied to Neon on deploy via `npm run build` and to PGlite preview via `src/lib/db.ts import.meta.glob`). Auth tables live in migrations/auth/0001_auth.sql and are only copied to migrations/0001_auth.sql when `VITE_AUTH_ENABLED=true`.

Overview



All tables are **unowned** (no `user_id`) — one shared plant. When auth is turned on, add `user_id TEXT NOT NULL` and scope every query with `requireUserId()` (`src/lib/auth/verify.server.ts`).

Tables

items — migrations/0002_plant.sql

Master of stocked spares.

COLUMN	TYPE	NOTES
id	serial PK	

COLUMN	TYPE	NOTES
code	text UNIQUE	e.g. HYD-040
name	text	Hydraulic Oil ISO 68
uom	text	Pcs / Ltr / Kg / Pair
reorder_level	int	Low threshold
qty	int	On-hand (mutated only via decideSlip / receiveSlip)
created_at	timestampz	now()

Indexes: code UNIQUE

machines — migrations/0002_plant.sql

COLUMN	TYPE	NOTES
id	serial PK	
code	text UNIQUE	CNC-01
name	text	CNC Lathe 01
line	text	Machine shop / Press bay

Seed: 6 machines (CNC-01, CNC-02, PRS-04, CNV-A, WLD-2, CMP-R).

slips — migrations/0002_plant.sql

Digital requisition. Replaces paper slip.

COLUMN	TYPE	NOTES
id	serial PK	
token	text UNIQUE	SLIP-2609-0001 (YYMM + seq)
item_id	int FK → items.id	NOT NULL
machine_id	int FK → machines.id	nullable
qty	int	requested
issued_qty	int	default 0
department	text	Production / Maintenance ...
station	text	Lathe cell

COLUMN	TYPE	NOTES
hod_title	text	Production HOD
hod_confirmed	bool	must be true
slip_date	date	
status	text	pending issued partial pr_open received
note	text	store note
created_at	timestampz	
decided_at	timestampz	nullable

Index: slips_status_idx (status, created_at desc)

movements — migrations/0002_plant.sql

Immutable ledger. **Negative qty = issue, positive = receive.**

COLUMN	TYPE	NOTES
id	serial PK	
item_id	int FK → items.id	
machine_id	int FK → machines.id	nullable
slip_id	int FK → slips.id	nullable (history seeds have no slip)
qty	int	-want for issue
kind	text	issue receive
created_at	timestampz	

Indexes: movements_item_idx (item_id, created_at desc), movements_machine_idx (machine_id, created_at desc)

Seed: 36 history movements over 58 days (issues & receives) to power 30d & 8-week charts.

bots — migrations/0003_bot_dashboard.sql

RPA definitions.

COLUMN	TYPE	NOTES
id	serial PK	

COLUMN	TYPE	NOTES
code	text UNIQUE	BOT-INV-SYNC
name	text	
description	text	
type	text	inventory slip reorder report generic
status	text	active paused error draft
cron_expr	text	* / 10 * * * * (nullable)
config	jsonb	per-bot settings, default {}
last_run_at	timestampz	
next_run_at	timestampz	
created_at	timestampz	
updated_at	timestampz	

Indexes: bots_status_idx , bots_type_idx **Seed:** 4 bots (Inventory Sync, Slip Processor, Reorder Watcher, Nightly Report).

bot_runs — migrations/0003_bot_dashboard.sql

Execution history.

COLUMN	TYPE	NOTES
id	serial PK	
bot_id	int FK → bots.id	ON DELETE CASCADE
status	text	running success failed skipped
trigger	text	schedule manual api webhook
started_at	timestampz	
finished_at	timestampz	
duration_ms	int	
input	jsonb	
output	jsonb	

COLUMN	TYPE	NOTES
error	text	nullable
created_at	timestamptz	

Indexes: bot_runs_bot_idx (bot_id, started_at desc) , bot_runs_status_idx **Seed:** 6×4 = 24 runs over last 7 days.

alerts — migrations/0003_bot_dashboard.sql

COLUMN	TYPE	NOTES
id	serial PK	
severity	text	info warn critical
title	text	
message	text	
bot_id	int FK → bots.id	nullable, SET NULL
item_id	int FK → items.id	nullable, SET NULL
acknowledged	bool	default false
created_at	timestamptz	

Indexes: alerts_severity_idx , alerts_ack_idx , alerts_bot_idx , alerts_item_idx **Seed:** Auto-seeded on first GET /api/alerts after items exist — one per item where qty <= reorder_level .

audit_logs — migrations/0003_bot_dashboard.sql

Append-only trail.

COLUMN	TYPE	NOTES
id	serial PK	
actor	text	system (or user email when auth on)
action	text	bot.create / slip.decide / alert.ack ...
entity_type	text	
entity_id	text	
payload	jsonb	

COLUMN	TYPE	NOTES
<code>created_at</code>	<code>timestampz</code>	

Indexes: `audit_logs_entity_idx` , `audit_logs_created_idx`

Auth (optional) — `migrations/auth/0001_auth.sql`

Better Auth `user` , `session` , `account` , `verification` (camelCase, quoted). Only active when copied to `migrations/0001_auth.sql` and `VITE_AUTH_ENABLED != "false"` .

Relationships

- `slips.item_id` → `items.id` (every slip is for one item)
- `slips.machine_id` → `machines.id` (which machine needs it)
- `movements.item_id` → `items.id` , `movements.machine_id` → `machines.id` , `movements.slip_id` → `slips.id`
- `bot_runs.bot_id` → `bots.id` (cascade)
- `alerts.bot_id` → `bots.id` (set null), `alerts.item_id` → `items.id` (set null)
- `audit_logs` is denormalized (no FK) for durability

Computed Views (not tables, via SQL)

- **InventoryRow:** `items` + 30d aggregates from `movements` + weekly buckets + `daysCover = qty / (consumed30/30)`
- **MachineStat:** per-machine `consumed30` , `distinctItems` , `topItem` , breakdown rows
- **Refill Dashboard:** `consumed30` vs `received30` , `receipts` , weekly series

Conventions

- **snake_case** in SQL, **camelCase** in TS (`plant.ts` / `bot.ts` mappers).
- **One migration file = one truth.** Never edit a shipped migration; add `0004_*.sql` .
- **PGLite parity:** `src/lib/db.ts` normalizes `int8→number` , `date→YYYY-MM-DD` , `interval→text` so preview == Neon.
- **Seed idempotency:** `ON CONFLICT (code) DO NOTHING` for bots/machines/items.
- **Indexes** on every FK + status + time for dashboard pagination.

How to add a new table

1. Create `migrations/0004_my_feature.sql` with `create table if not exists ...`
2. Run `npm run build` (also runs `db:migrate` for Neon) or just restart dev (PGLite auto-applies on HMR).
3. Add types in `src/lib/my_feature.ts` , server fns in `src/lib/my_feature-server.ts` , and REST in `src/routes/api/...` + `server/routes/api/...` .

4. Document here and in `/system` ER.

ER Diagram (Mermaid)

```
erDiagram
    items ||--o{ slips : "item_id"
    machines ||--o{ slips : "machine_id"
    items ||--o{ movements : "item_id"
    machines ||--o{ movements : "machine_id"
    slips ||--o{ movements : "slip_id"
    items ||--o{ alerts : "item_id"
    bots ||--o{ bot_runs : "bot_id"
    bots ||--o{ alerts : "bot_id"

    items {
        serial id PK
        text code UK
        text name
        text uom
        int reorder_level
        int qty
    }
    machines {
        serial id PK
        text code UK
        text name
        text line
    }
    slips {
        serial id PK
        text token UK
        int item_id FK
        int machine_id FK
        int qty
        int issued_qty
        text status
    }
    movements {
        serial id PK
        int item_id FK
        int qty
        text kind
    }
    bots {
        serial id PK
        text code UK
        text status
        text cron_expr
        jsonb config
    }
    bot_runs {
        serial id PK
        int bot_id FK
    }
```

```
    text status
    text trigger
  }
  alerts {
    serial id PK
    text severity
    bool acknowledged
  }
```